Fixed and Moving Local References in the Rotemberg–HSA Phillips Curve

Restriction Version: F0, U, R1, R2, and R3

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Automatically built from provenance-validated production runs

Abstract

This restriction-version report is separate from `nkpc_hsa_report.pdf`, which remains the historical reduced-form and ablation report. The present document compares a fixed-reference local Rotemberg–HSA benchmark (F0) with an unrestricted moving-reference model (U), the local HSA cross-restricted model (R1), its weak Third-law extension (R2), and a constant-pass-through local benchmark (R3). Every numerical table and figure is generated only from the signed theory-run namespace. Historical artifacts are never substituted for a missing current run.

Contents

1 Research question and evidence ladder

The empirical design separates four logically different objects: fixed-reference theory identities, maintained HSA laws, domain and local-model admissibility, and the moving-reference approximation. The evidence ladder is

$$\text{fixed-reference theory} \to F0 \to \text{moving reference} \to U \to R1 \to R2, \qquad R3 = R1 + \{\gamma = 0\}.$$

F0 is not nested in U/R1/R2/R3. Within the moving-reference family, $U \supset R1 \supset R2$. R3 is an equality-restricted local benchmark and is not described as a full Co-PaTh model.

2 Current research design: fixed and moving local references

The current specification starts from the quarterly, fixed-reference local Rotemberg–HSA equation. F0 is its fixed-N_0 empirical benchmark. U, R1, R2, and R3 add a slow-moving local reference; this moving-reference step is a semi-structural approximation and is not Eq. 21 itself.

2.1 Restriction taxonomy

Category	Contents
Theory identity / cross-equation relation	kappa0_structural = (zeta0 - 1) / chi; Theta0 = (1 / chi) * (1 - rho0) / rho0; mu0 = zeta0 / (zeta0 - 1); d kappa / d log N = zeta * Theta
Maintained HSA law	Second law: theta0 > 0 and kappa_N_empirical > 0; Weak Third law: gamma <= 0 (R2 only)
Domain / local admissibility	chi > 0; zeta0 > 1; mu0 > 1; kappa0 > 0; optional whole-path kappa_t > 0 and theta_t > 0
Moving-reference empirical approximation	moving Nstar_t is a slow-moving local reference; kappa_t = kappa0 + kappa_N_empirical * Nbar_t; theta_t = theta0 + gamma * Nbar_t
Deliberately not imposed (future R4)	kappa0 = (zeta0 - 1) / chi; theta0 = (10 / chi) * (1 - rho0) / rho0

2.2 Model hierarchy and non-nesting

Within the moving-reference family, $U \supset R1 \supset R2$ and R3 is $R1 + \{\gamma = 0\}$. F0 is not nested in that family because its reference is fixed.

Model	Slug	Reference	Cross restriction	Second law	Third law	Gamma	State sampler
F0	hsa_f0	fixed reference	no	maintained	neither	not_applicable	fixed reference ffbs
U	hsa_u	moving reference	no	not maintained	neither	unrestricted	particle gibbs
R1	hsa_r1	moving reference	yes	maintained	neither	unrestricted	particle gibbs
R2	hsa_r2	moving reference	yes	maintained	weak	gamma <= 0	particle gibbs
R3	hsa_r3	moving reference	yes	maintained	neither	gamma = 0	joint ffbs

2.3 Quarterly structural scaling

With $\bar{N} = 10(\log N - \log N_0)$ and inflation in percentage points, $d\kappa/d\log N = 10\kappa_N$ and the restricted marginal-cost specification has $100\kappa_N = b_x\zeta_0\theta_0$. The equality is never imposed directly on a gap proxy without an explicit b_x, or on the legacy direct 4Q-YoY regression.

2.4 Current posterior results

Model	Parameter	Mean	SD	2.5%	97.5%	Convergence
F0	kappa_0	0.0852	0.0608	0.0042	0.2264	converged
F0	theta_0	0.0446	0.0244	0.0044	0.0963	converged
U	kappa_0	0.0919	0.0540	0.0149	0.2211	converged
U	kappa_N_empirical	0.0006	0.0146	-0.0291	0.0297	converged
U	d_kappa_d_logN	0.0059	0.1455	-0.2907	0.2970	converged
U	theta_0	0.1261	0.0827	0.0203	0.3378	converged
U	gamma	-0.0033	0.0169	-0.0380	0.0297	converged
R1	kappa_0	0.0846	0.0531	0.0151	0.2136	converged
R1	kappa_N_empirical	0.0068	0.0043	0.0011	0.0178	converged
R1	d_kappa_d_logN	0.0678	0.0430	0.0110	0.1783	converged
R1	theta_0	0.1129	0.0716	0.0183	0.2972	converged
R1	gamma	-0.0033	0.0166	-0.0370	0.0291	converged
R2	kappa_0	0.0844	0.0530	0.0152	0.2135	converged
R2	kappa_N_empirical	0.0067	0.0042	0.0011	0.0171	converged
R2	d_kappa_d_logN	0.0673	0.0419	0.0112	0.1713	converged
R2	theta_0	0.1122	0.0698	0.0187	0.2856	converged
R2	gamma	-0.0143	0.0111	-0.0413	-0.0005	converged
R3	kappa_0	0.0789	0.0547	0.0085	0.2080	converged
R3	kappa_N_empirical	0.0036	0.0031	0.0001	0.0115	converged
R3	d_kappa_d_logN	0.0359	0.0307	0.0015	0.1152	converged
R3	theta_0	0.0599	0.0511	0.0025	0.1921	converged

2.5 Local-model admissibility diagnostics

The Gaussian probe column reports how often an unconstrained conditional coefficient draw would violate the imposed admissible region; it does not enter the posterior. High probe pressure is reported as a diagnostic that the local linear approximation or posterior geometry deserves review. State rejection is the share of proposed latent paths rejected by whole-path positivity.

Model	Probe nonadmissible %	State reject %	Stored kappa<=0 %	Stored theta<=0 %	Local-range flag
F0	–	–	0.0000	0.0000	none
U	66.5625	1.6416	0.0000	0.0000	review
R1	62.9569	1.4316	0.0000	0.0000	review
R2	78.3520	1.1239	0.0000	0.0000	review
R3	55.0403	0.6333	0.0000	0.0000	review

2.6 Q/Q versus four-quarter YoY observation design

The two transformations of a common price index are estimated in separate likelihoods and are never stacked as if independent. F0 and U provide the direct observation-design comparison. R1–R3 remain Q/Q-only in this revision because a direct YoY regression does not preserve the meaning of the quarterly structural cross-equation coefficient.

Model	Inflation	T	Status	kappa0	theta0	kappaN
F0	Q/Q	124	converged	0.0852	0.0446	–
U	Q/Q	124	converged	0.0919	0.1261	0.0006
F0	4Q YoY	124	converged	0.1169	0.0747	–
U	4Q YoY	124	converged	0.1159	0.1283	-0.0016

2.7 Current-run provenance

Estimation revision: `2026-08-moving-reference-hsa-v1`; estimation code: `204a2f319b88`; sample: 1982-03-31–2012-12-31; report builder code: `16d7921e29eb`; builder dirty: no.

Model	Run ID	Inflation	T	Status
F0	20260810_175459	qoq	124	converged
R1	20260810_175459	qoq	124	converged
R2	20260810_175459	qoq	124	converged
R3	20260810_175657	qoq	124	converged
U	20260810_175459	qoq	124	converged
F0	20260810_180342	yoy_4q	124	converged
U	20260810_180540	yoy_4q	124	converged

2.8 Restriction-relevant posterior figures

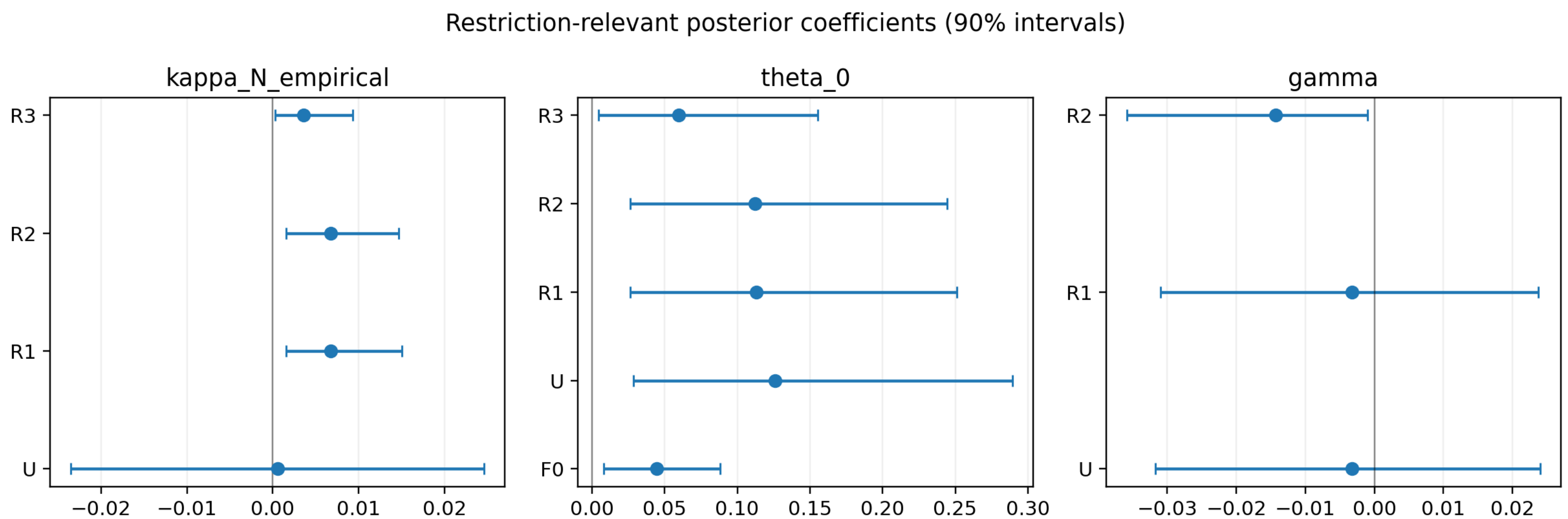

Figure 1: Restriction-relevant posterior coefficients.

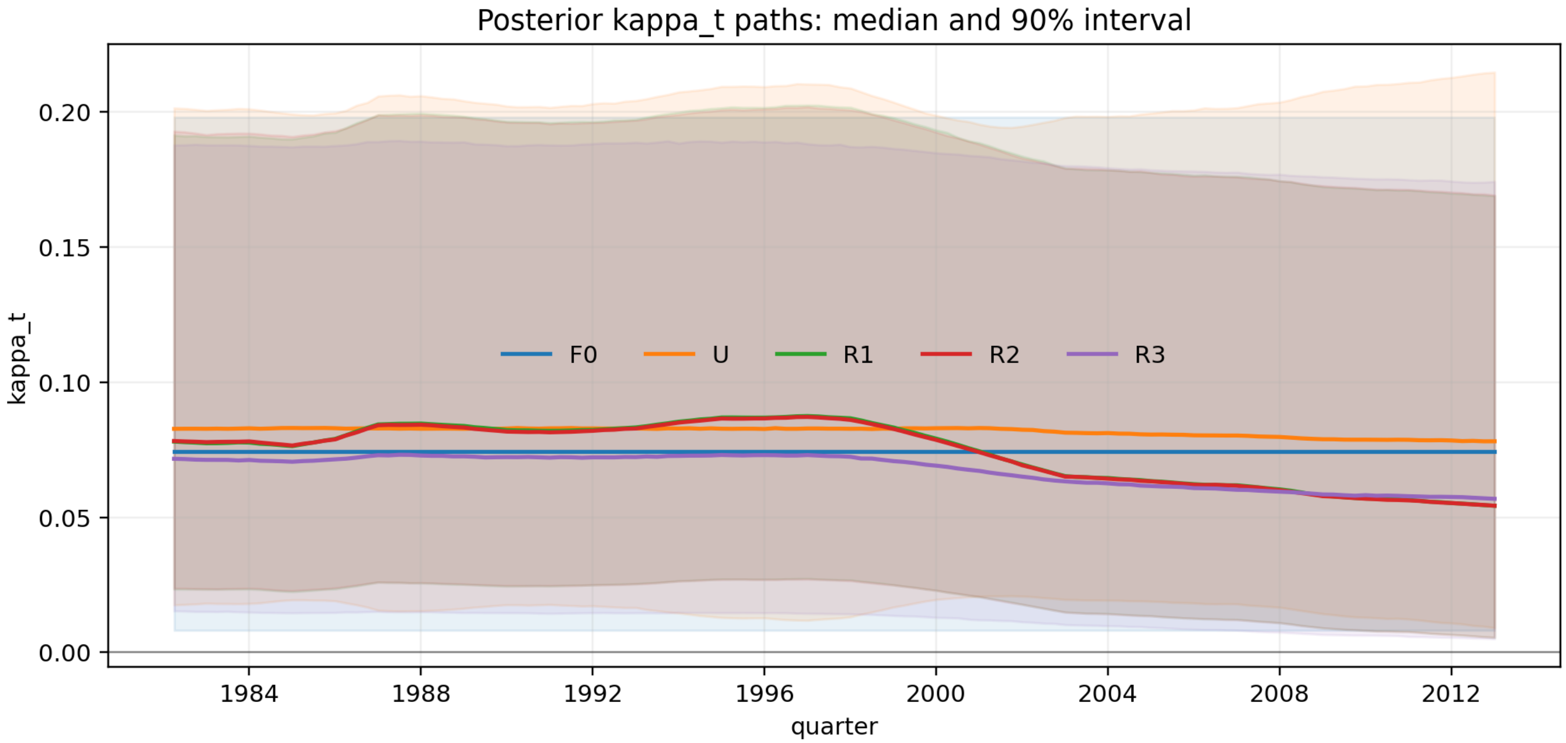

Figure 2: Posterior slope paths by model.

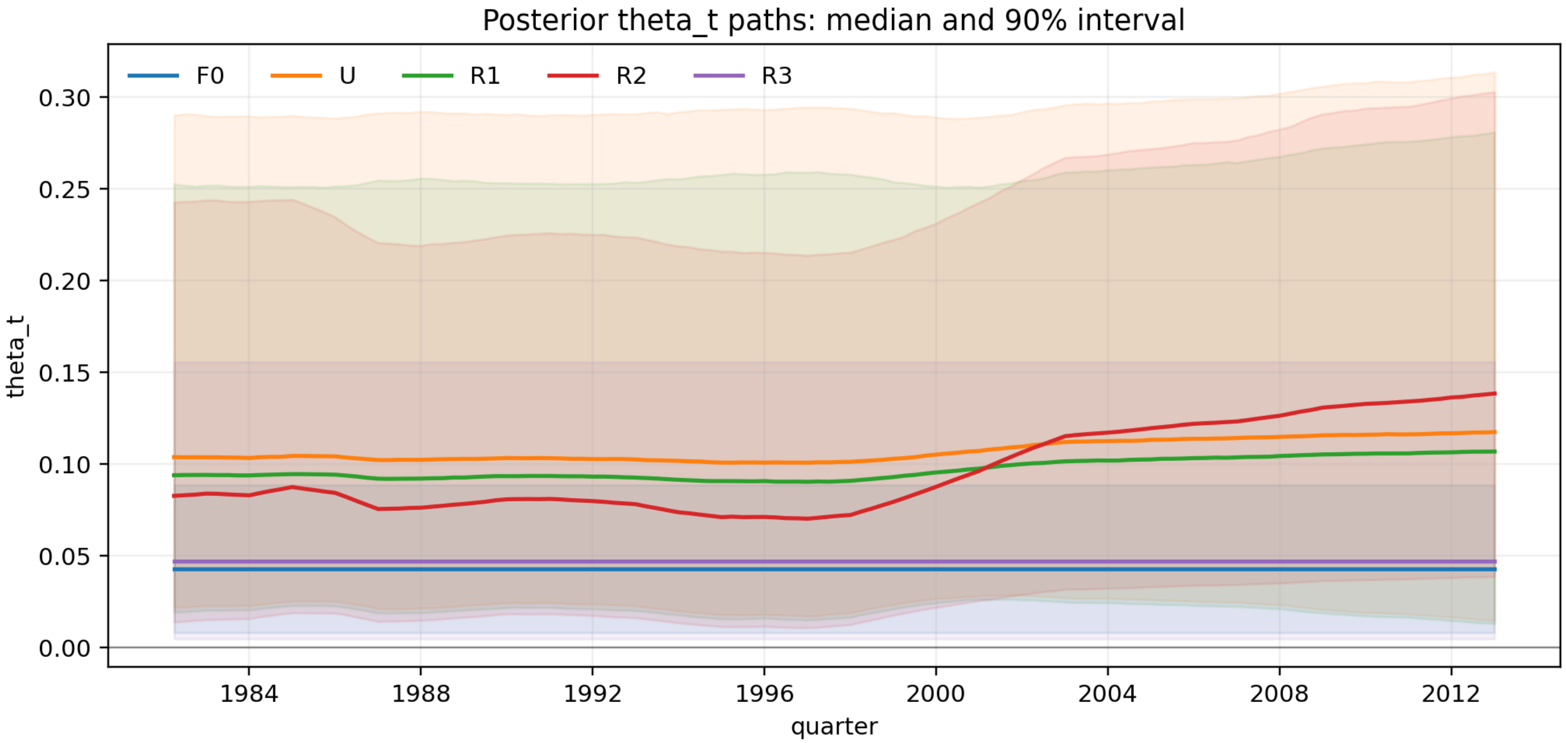

Figure 3: Posterior entry-coefficient paths by model.

3 Interpretation of nested comparisons

The $U \to R1$ comparison isolates tension with the local HSA cross-equation restriction. The $R1 \to R2$ comparison isolates tension with the weak Third law, while $R1 \to R3$ assesses whether locally constant pass-through is too restrictive. A restriction is not presumed to improve fit. Posterior geometry, convergence, admissibility rejection, and fit must be read together.

4 Observation-design discipline

Quarterly structural frequency is fixed in every run. Q/Q and four-quarter YoY are alternative observation designs and are never stacked in one likelihood. The current direct YoY regression is available only as an F0/U comparison; R1–R3 remain Q/Q-only until a likelihood that aggregates four quarterly structural equations is implemented. Thus the quarterly restriction $100\kappa_N = b_x\zeta_0\theta_0$ is never imposed directly on a reduced-form four-quarter coefficient.

5 Reproducibility and stale-artifact policy

The run manifest records code revision, model definition and hierarchy, exact restrictions, maintained laws, admissibility conditions, inflation and expectation transformations, activity and competition proxies, N_0, ζ_0/μ_0 treatment, sample, sampler, chains, seeds, and convergence. The report build fails if a comparison would mix incompatible revisions, samples, transformations, observation designs, proxies, priors, or model definitions. The historical report and its result directory are not scanned by this builder.